DO NOW — RECALL

Lesson 135 · Entity Relationships: One-to-Many and Mar

Name: _____ Class: _____ Date: _____

1. What is the role of a foreign key?

2. In Big O notation, what is binary search worst-case complexity?

3. What does ROM stand for?

4. Which is removed first from a FIFO queue?

DO NOW — CHECK / OBJECTIVES

Correct and improve

Check page 1 in GREEN PEN. Write corrections in full.

1. Model answer

It references a candidate/primary key in a related table.

2. Model answer

O(log n) for sorted random-access data.

3. Model answer

Read-only memory.

4. Model answer

The longest-waiting item at the front.

By the end, you can…

• Find objects with independent identity, not every noun.

• One student takes many courses, and each course has many students.

• Add Enrolment table referencing StudentID and CourseID.

Vocabulary — know these terms

entity — An identifiable class of thing stored in database.

relationship — Association between entities.

cardinality — How many related instances are possible.

junction table — Table breaking a many-to-many relation into two one-to-many relations.

NEW LEARNING

Read, annotate and explain

Identify entities

Find objects with independent identity, not every noun.

Set cardinality

One student takes many courses, and each course has many students.

Resolve M:N

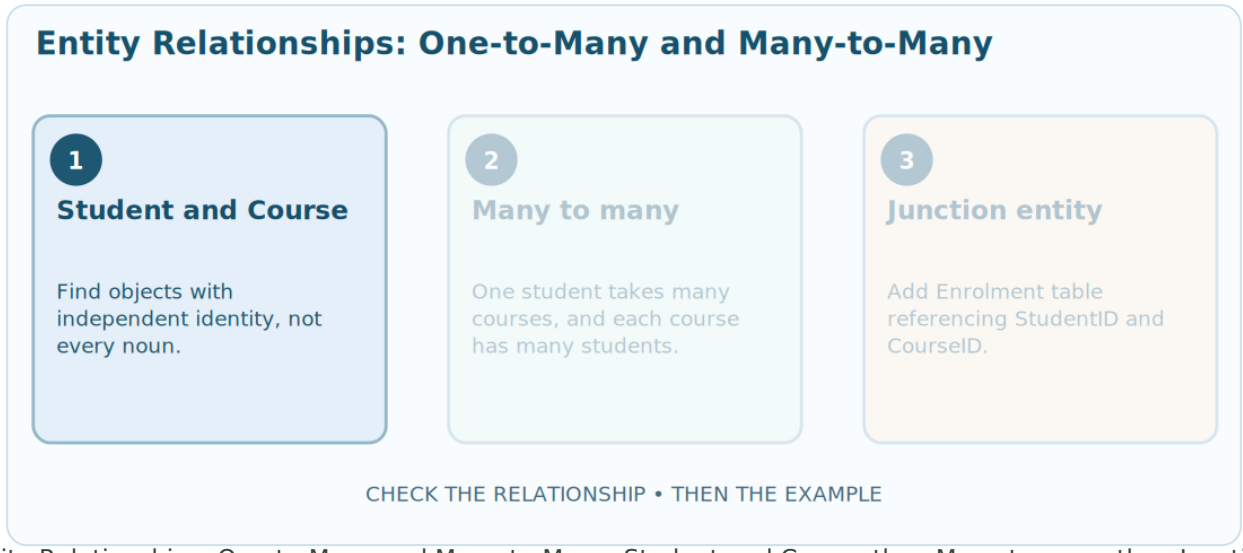
Add Enrolment table referencing StudentID and CourseID.

Explain one step in your own words, using the correct technical terms.

New Learning · original lesson diagrams

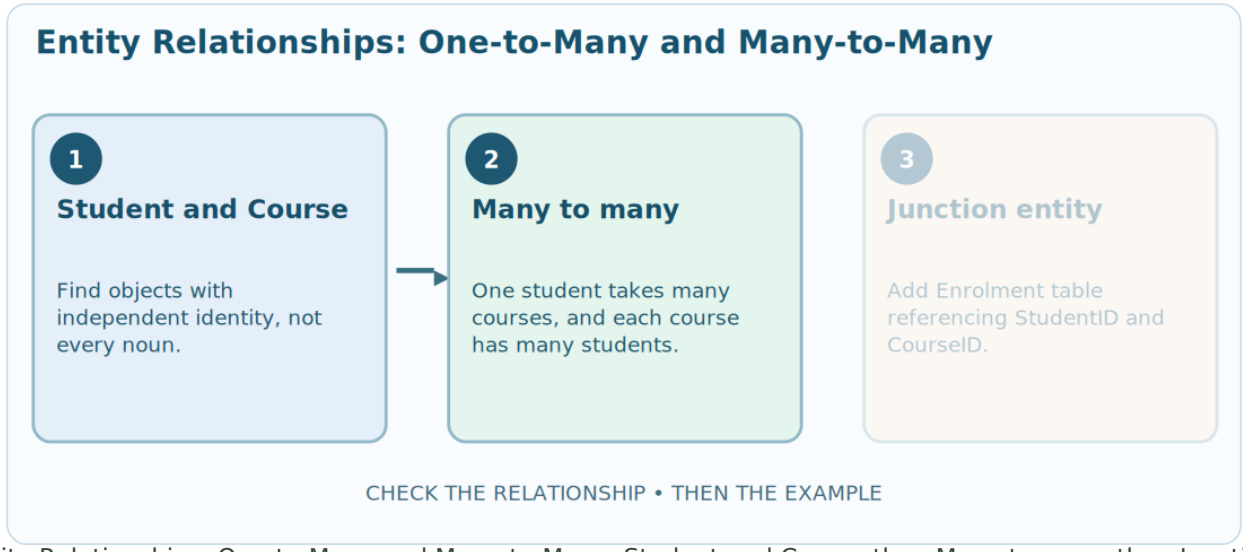
Diagrams appear after the related explanation. Annotate the original source visual.

New Learning



Entity Relationships: One-to-Many and Many-to-Many; Student and Course then Many to many then Junction entity

New Learning

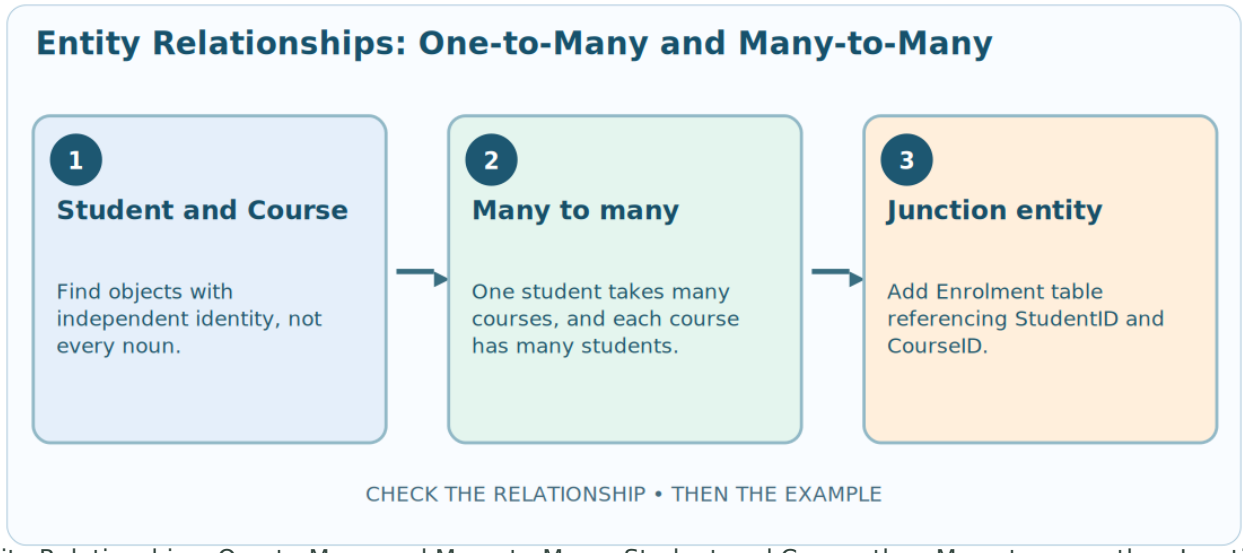


Entity Relationships: One-to-Many and Many-to-Many; Student and Course then Many to many then Junction entity

New Learning · original lesson diagrams

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New Learning



Entity Relationships: One-to-Many and Many-to-Many; Student and Course then Many to many then Junction entity

I DO — WORKED EXAMPLE

Follow the worked method

Represent students enrolling on several courses, with many students per course.

Student↔Course is M:N

Find the given data and name the required result.

Create Enrolment bridge

Explain why this intermediate stage is necessary.

Two one-to-many links

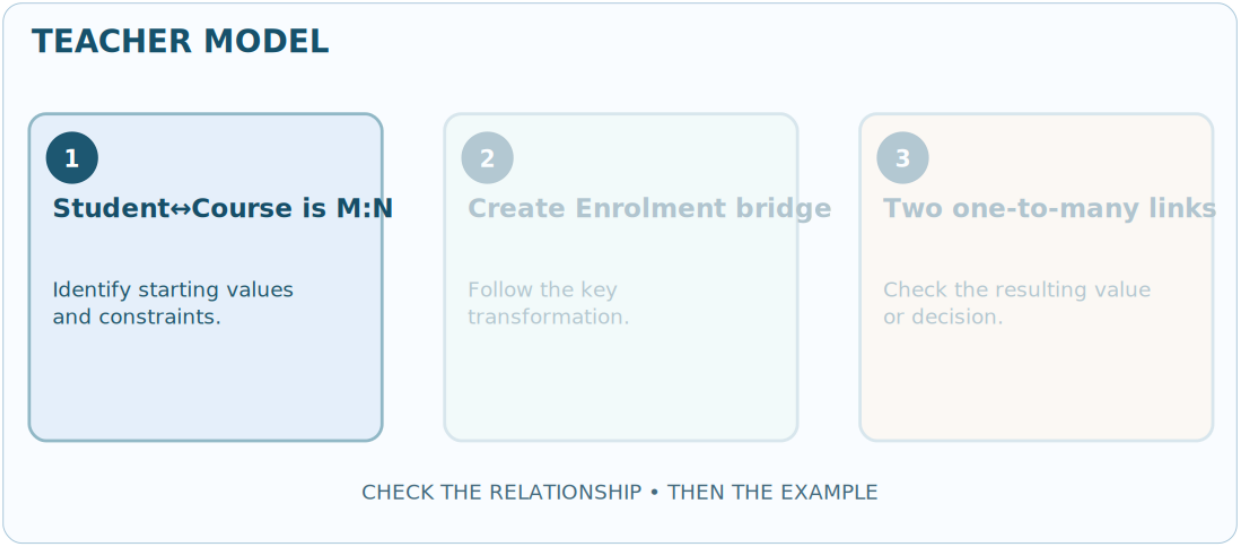
Student(StudentID PK) and Course(CourseID PK) form many-to-many. Resolve with Enrolment(StudentID FK, CourseID FK, EnrolledOn), with composite PK (StudentID, CourseID) if enrolment is unique per pair.

I Do · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

Represent students enrolling on several courses, with many students per course.

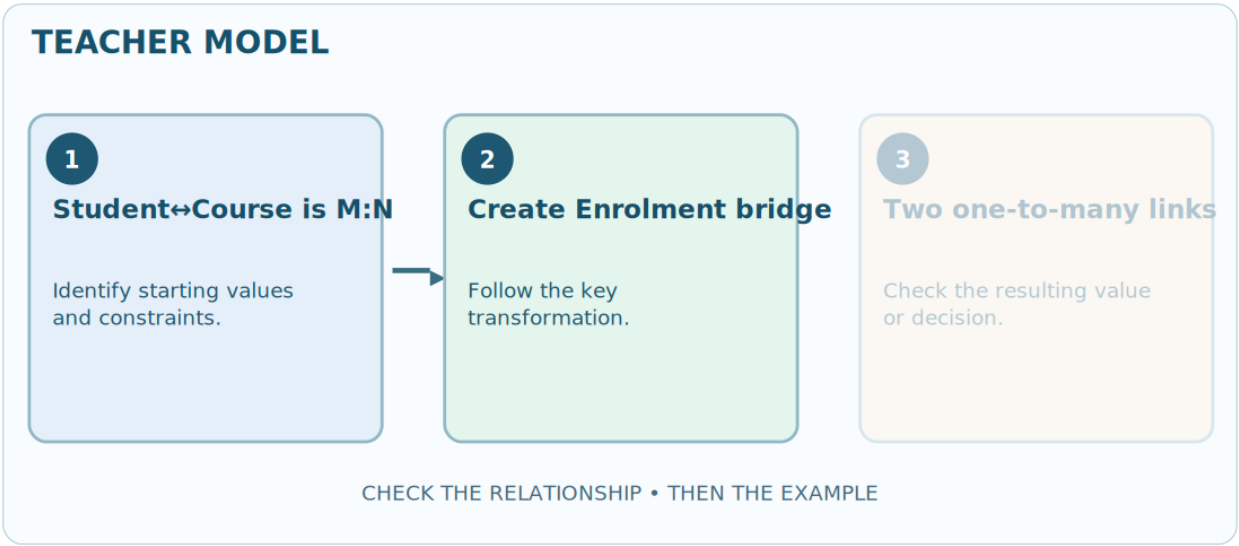
Find the given data and name the required result.



TEACHER MODEL; Student↔Course is M:N then Create Enrolment bridge then Two one-to-many links

Represent students enrolling on several courses, with many students per course.

Find the given data and name the required result.



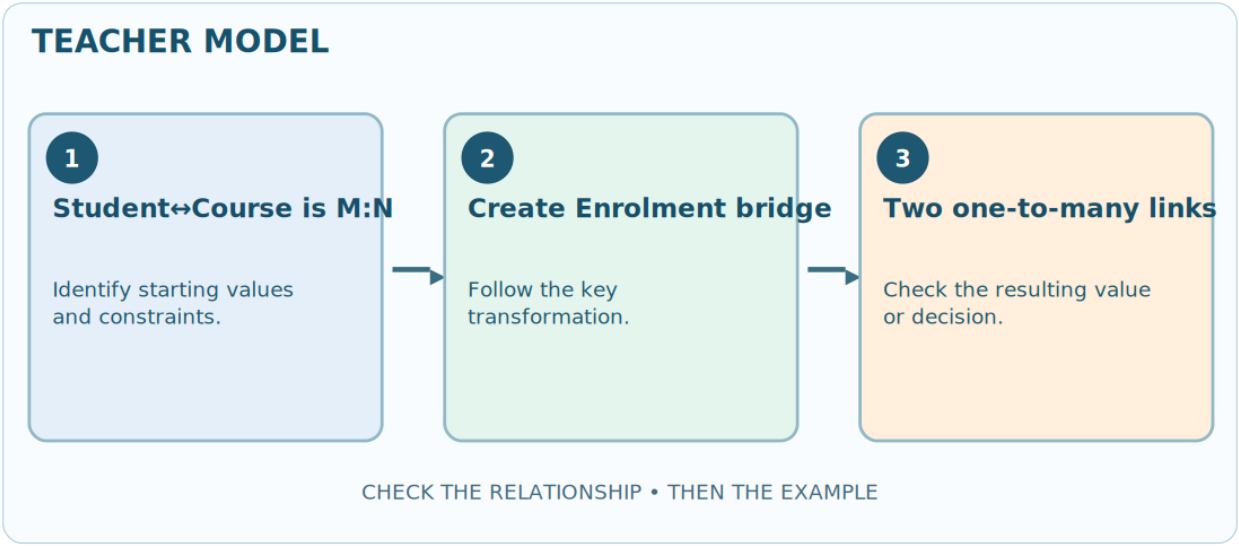
TEACHER MODEL; Student↔Course is M:N then Create Enrolment bridge then Two one-to-many links

I Do · original lesson diagrams

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WE DO — GUIDED PRACTICE

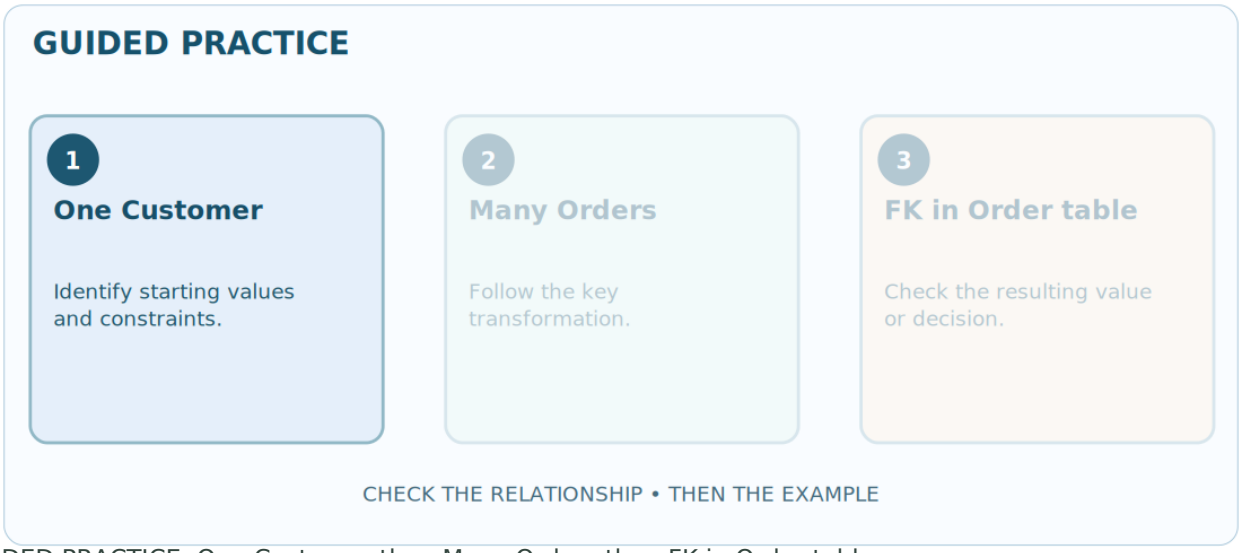
Try → Discuss → Improve

We Do · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

A customer can place multiple orders; an order belongs to one customer. Model the relationship.

Find the given data and name the required result.



GUIDED PRACTICE; One Customer then Many Orders then FK in Order table

WE DO — TRY · DISCUSS · IMPROVE

TRY: Use the original guided questions and diagrams in the preceding We Do pages. DISCUSS: Explain and challenge methods before independent practice. IMPROVE: Note one change now; compare with the source model only AFTER the Exit Ticket.

Guided 1: One Customer Find the given data and name the required result.

DISCUSS — Compare your methods. Justify one important decision, identify one difference, and agree a specific improvement.

IMPROVE — Write one precise correction or stronger explanation in your own answer.

YOU DO — INDEPENDENT PRACTICE

Questions · Part 1

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 1

Explain the difference between one-to-many and many-to-many relations with examples. [6]

Question 2

Design tables for Author and Book when each book has multiple authors. [6]

YOU DO — INDEPENDENT PRACTICE

Questions · Part 2

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 3

Evaluate why a field containing comma-separated ClubIDs is poorer than a Membership junction table. [8]

EXIT TICKET / REFLECTION

Check your understanding

Complete the actual lesson Exit Ticket before checking the answer.

A many-to-many relationship can always be represented correctly with just a single FK on one of the two entity tables.

How confident are you? Circle one: Confident / Getting there / Need help

Explain your choice, identifying one concept you can now use independently.

GREEN PEN — Correct and explain

Choose one mistake from your independent work. Write the correct method and explain precisely why your original answer was wrong.

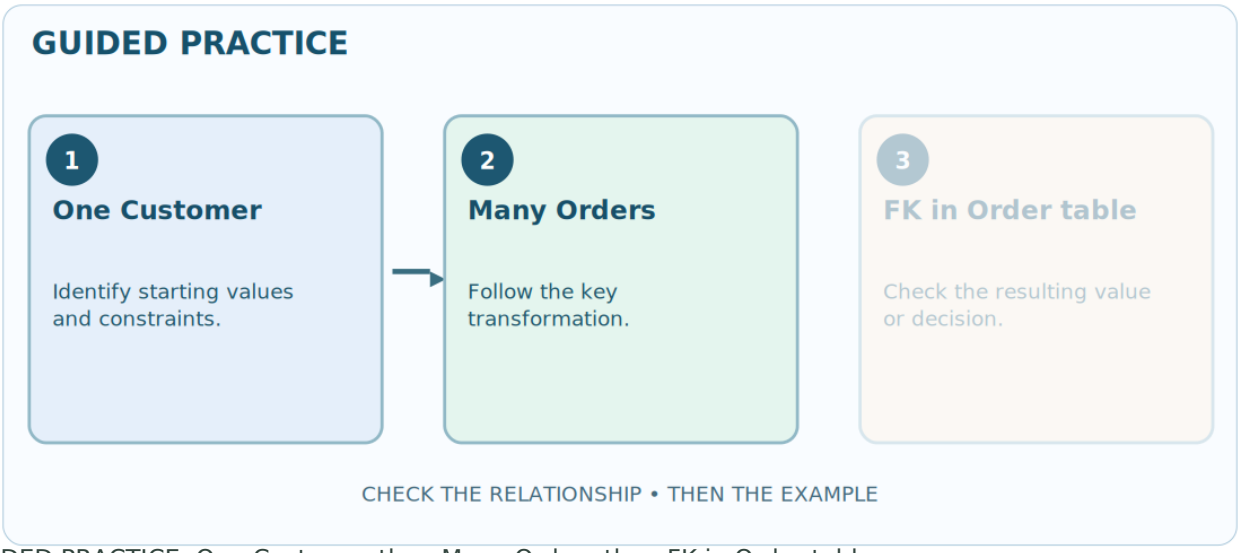
Next step: revisit the model or a relevant revision-guide section if needed.

We Do Check · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

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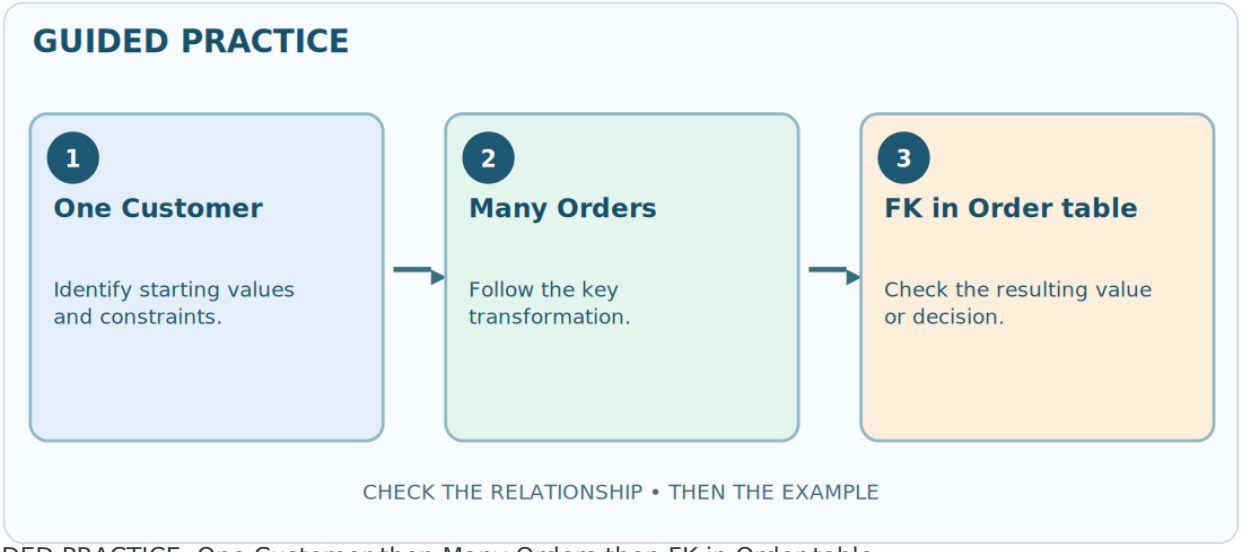
Find the given data and name the required result.



GUIDED PRACTICE; One Customer then Many Orders then FK in Order table

A customer can place multiple orders; an order belongs to one customer. Model the relationship.

Find the given data and name the required result.



GUIDED PRACTICE; One Customer then Many Orders then FK in Order table

WE DO — SOURCE CHECK / IMPROVE

Only use this section AFTER attempting the We Do, You Do and Exit Ticket. Compare with the original classroom lesson's CHECK explanation. Correct errors in GREEN PEN and explain why.

Guided 1 — original CHECK

Many Orders

Explain why this intermediate stage is necessary.

FK in Order table

Customer 1:M Order. CustomerID PK in Customer; CustomerID FK in Order. OrderID serves as the Order PK. Optionality (must a customer have an order?) needs a business rule.

IMPROVE: Explain one correction to your own reasoning.